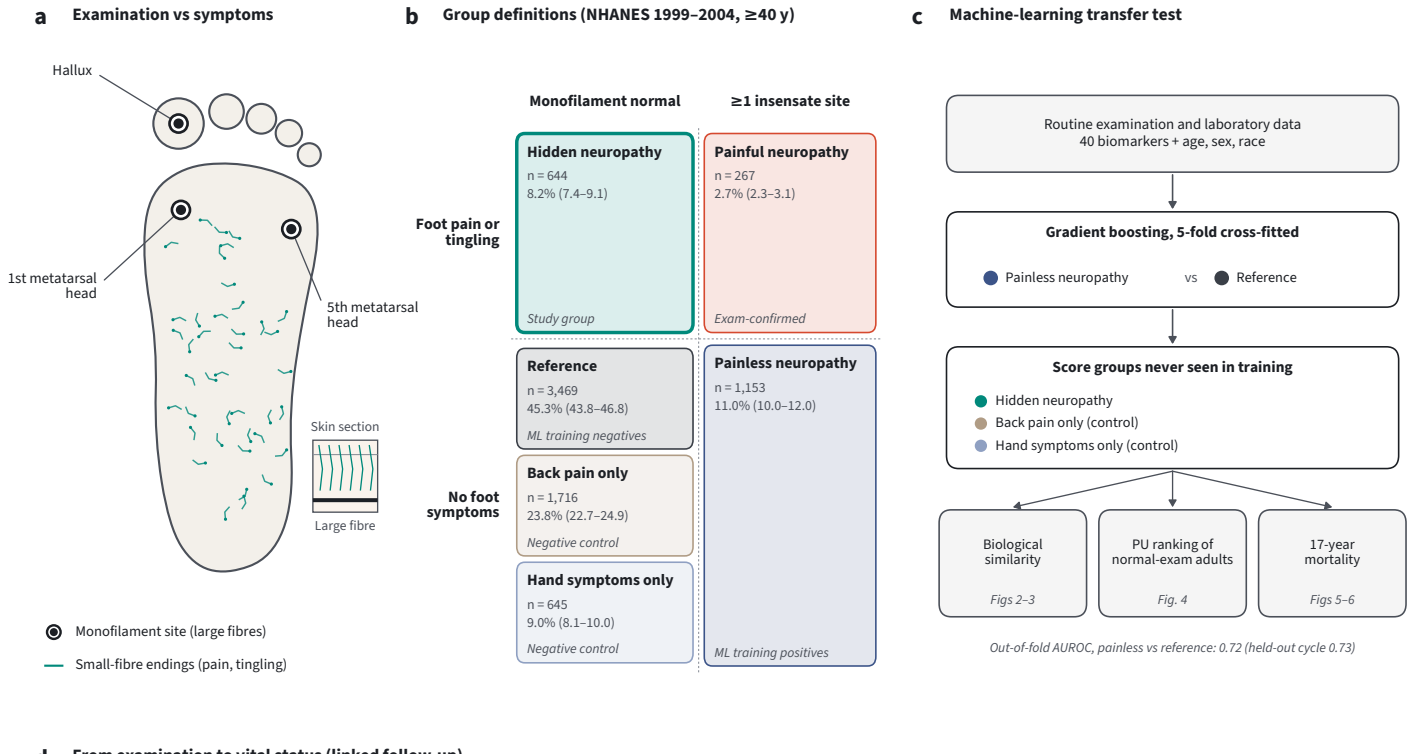
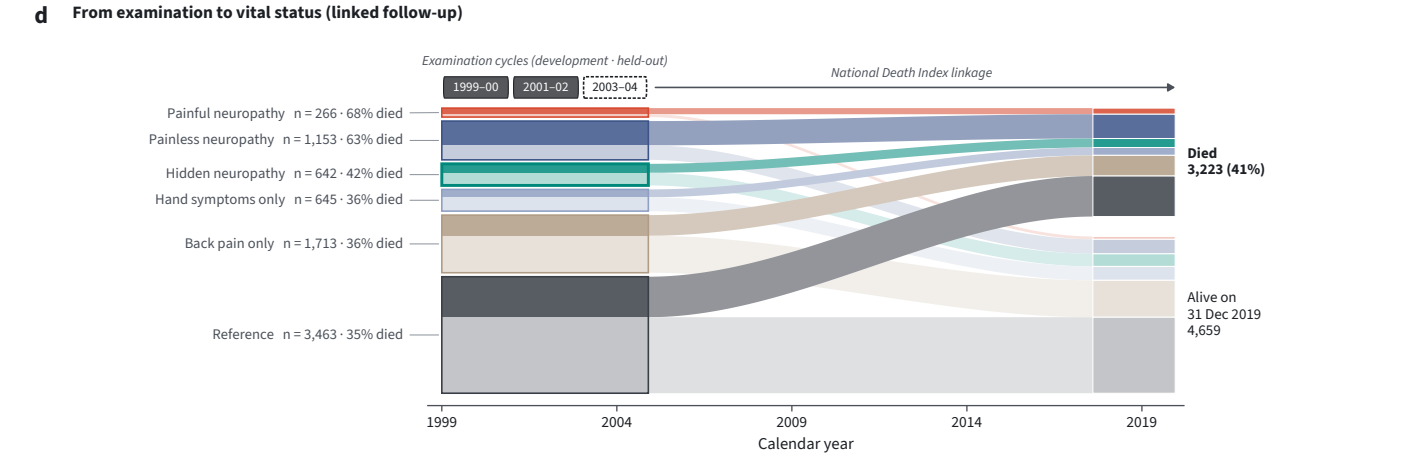

a
Examination vs symptoms
Hallux
1st metatarsal head
5th metatarsal head
Skin section
Large fibre
Monofilament site (large fibres)
Small-fibre endings (pain, tingling)
b
Group definitions (NHANES 1999–2004, ≥40 y)
Monofilament normal
≥1 insensate site
Foot pain or tingling
No foot symptoms
Hidden neuropathy
n = 644
8.2% (7.4–9.1)
Study group
Painful neuropathy
n = 267
2.7% (2.3–3.1)
Exam-confirmed
Reference
n = 3,469
45.3% (43.8–46.8)
ML training negatives
Painless neuropathy
n = 1,153
11.0% (10.0–12.0)
Back pain only
n = 1,716
23.8% (22.7–24.9)
Negative control
Hand symptoms only
n = 645
9.0% (8.1–10.0)
Negative control
ML training positives
c
Machine-learning transfer test
Routine examination and laboratory data
40 biomarkers + age, sex, race
Gradient boosting, 5-fold cross-fitted
Painless neuropathy
vs
Reference
Score groups never seen in training
Hidden neuropathy
Back pain only (control)
Hand symptoms only (control)
Biological similarity
Figs 2–3
PU ranking of normal-exam adults
Fig. 4
17-year mortality
Figs 5–6
Out-of-fold AUROC, painless vs reference: 0.72 (held-out cycle 0.73)


d
From examination to vital status (linked follow-up)
Examination cycles (development · held-out)
1999–00
2001–02
2003–04
National Death Index linkage
Painful neuropathy n = 266 · 68% died
Painless neuropathy n = 1,153 · 63% died
Hidden neuropathy n = 642 · 42% died
Hand symptoms only n = 645 · 36% died
Back pain only n = 1,713 · 36% died
Reference n = 3,463 · 35% died
Died
3,223 (41%)
Alive on
31 Dec 2019
4,659
1999
2004
2009
2014
2019
Calendar year